2E-EDUC 6 - FS Field Study

Term 2 AY 2026 - 2026

Worksheet: Critique, Reflections on a Sample Classroom-Based Action Research Report and CBAR Title Proposals

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Part 1. Familiarization with the Different Sections of a Classroom-based Action Research Report and Critique for Improvement

After reading the full report of a sample Classroom-based Action Research found below entitled "The Use of Contextualized Video Instruction in Science 4 Pupils' Academic Achievement", please accomplish this table to guide you through the sections of an action research report.

Section/Elements	What is your understanding of this section? What purpose do you think does it serve?	What are your comments and/or suggestions to improve this section as written by the authors?
Title	The title shows the variables involved in the studies and the target population. It also shows how a variable (Contextualized Video Instruction) is being used to find an observed change (academic achievement) following the implementation of the said variable.	We can drop the "The Use of ..." as it can already be implied with the remaining parts of the title.
Abstract	It shows an overview of the entire study.	It is already well-presented.
Keywords	It serves as a way for people who access journals for the paper to be recommended in case when they are searching for specific concepts, themes, etc.	It is already encompassing.
1. Introduction	It shows the context of how and why the researchers has shown interest in implementing contextualized video instructions to improve academic achievment.	None. The introduction has thoroughly established the need for the study.
1.1 Statement of the Problem	List of the aims as to why the research is being conducted.	It fits well with the quasi-experimental nature of the research.
1.2 Review of Related	It shows supporting evidences	It is well written and

Literature and Studies	of previously conducted researches and studies that guides the actual CBAR so it wouldn't be too way off from the topic at hand.	the citations are curated with intent rather than simply citing related literature and studies.
1.3 Theoretical Framework	It features Edgar Dale's Cone of Experience to explain how different levels of experience lead to understanding.	It is already well presented.
1.4 Conceptual Framework	It uses IPO model to show the process and the to-be-proposed action plan.	It is already well presented .
1.5 Limitations and Delimitations	It shows the scope of the study, especially the limits of the variables and explain the possible extraneous variables that are beyond the research's capability to control.	It is already well presented.
1.6 Definition of Terms	It features the different terms conceptually and operationally defined in the study.	It is already well presented.
2. Methods	It explains the highly quasi-experimental pretest-posttest design of the CBAR.	It is already well presented.
2.1 Research Design	It identifies that the study as a quasi-experiment which aims to measure the impact of the contextualized video instruction.	It is already well presented.
2.2 Research Locale	It describes the setting where the CBAR will be implemented.	It could be improved by giving some context to the available technological capabilities of the school.
2.3 Respondents of the Study	It shows the profile of the participants of the study.	It is already well presented.
2.4 Instrumentation	It lists the tools used to extract information from the respondents, such as the questionnaire, and the exams as well as the rubrics for grading the exams.	It is already well presented. The instrument is appended at the end of the paper.
2.5 Data Gathering Procedure	Details the timeline and the process of disseminating the tests via Google Forms	It would be better if some flowchart was added.
2.6 Statistical Treatment of Data	It shows how the raw data was transformed into a meaningful data which answers the goals of the study.	It is already well-presented.
2.7 Ethical Considerations	It shows the researchers' thinking process when dealing with sensitive privacy, confidentiality, and no harm	A fruther discussion on how the pretest and posttest was administered. Were

	intentions to the participants.	there opportunities for the students to search online for answers?
3. Results and Discussions	It shows the findings from the study after discussing how the data was extracted.	It is already well-presented.
Note: Please describe in reference to the Research Questions.		
4. Conclusions and Recommendations	It now shows the summary of results and discussion and how it supports or debunks the existing notions of the highlighted study as supported by the RRLS.	It is already well-presented.
4.1 Conclusions	Contextualized videos are effective tools for asynchronous learning and improve academic attainment, at least in Araling Panlipunan on Science 4 pupils who participated in the CBAR.Contextualized videos are effective tools for asynchronous learning and improve academic attainment.	It is already well-presented.
4.2 Recommendations	Faculty should continue using these resources; future researchers should use control groups .	It is already well-presented.
Acknowledgement	The researcher's section where they thank those who has been helpful in their study.	It is already well-presented.
References	It lists all consulted works in the CBAR.	It is already well-presented.
Appendices	It contains the materials that was consolidated from the study.	It is already well-presented.
Appendix A- Learners' Profile	List all students who participated in the CBAR.	It is already well-presented.
Appendix B – Validated Teacher-Made Test Evaluation	It shows the exact test that has underwent Content Validation by subject matter experts and was actually used as a way to collect data to answer the research goals.	It is already well-presented.
Appendix C – Table of Specifications and 50-Item Pre-test and Post Test	It shows the detailed breakdown of the pretest and posttest item as well as the table of specification which serves as the detailed guide of how the data fits to answer the research questions.	It is already well-presented.

Note: The sample CBAR report was written by Stephen Allen M. Dy and Clarisa May D. Gabion, Practice Teachers (Faculty of Science, Technology and Mathematics, Institute of Teaching and

Learning, Philippine Normal University-Manila).

Part II. My Reflection on the Action Research Report

An essential function of the action research project is to value the importance of looking back on one's work, both in terms of process and results. The overarching questions to answer are **'How do you intend to use the results of this project?'** and **'What would I do differently next time?'**

Hypothetically, you assume the role of a researcher of the classroom -based action research you just read and critiqued.

The following prompts can help you formulate a purposeful reflection on this research project:

1. Given the results, how will you use this information to:

    a) improve student learning in the future?

    I will adopt these contextualized video instructions to my class. I will make use of LLMs like Google NotebookLM to create video instructions that fits my learning needs for my class. I will also using signaling (emphasizing keywords) and chunking lengthy topics into manageable ideas.

    b) improve your teaching practice?

    Using these contextualized video instructions will help me become more student-centered approach given that the video instructions will be tailored to the needs of the students.

2. What additional questions arose that you might want to pursue?

    Does using this contextualized video instruction applies to subjects other than science? I would love to use these practices into Social Science.

3. How might you change the research questions in light of the results?

    I would make a follow-up study adopting a Between Subjects Designs (groupings) so that it can address limits of a Within Subjects Design (pretest-postttest). I would also try to find AI tools for me to check the essay outputs more efficiently in order to reduce workload and tend to other teacher obligations.

4. What other innovation might you try next time, based on these results?

    Please write down at least 2 proposed research titles and at least 2 possible research questions for each proposed title.

Enhancing Environmental Literacy through a Choice-Based Consequence Simulation Game in Grade 4 Science	Utilizing Interactive Virtual Optic Labs to Improve Conceptual Understanding of the Properties of Light
What is the level of students' environmental literacy before and	What are the common misconceptions of students regarding how light

after the intervention?	travels?
How do students perceive the cause-and-effect relationships within the interactive game?	To what extent does granular control of variables in the app improve student engagement?
Is there a significant difference in the students' ability to predict environmental outcomes after playing the simulation?	How does the use of virtual optic labs compare to traditional illustrations in achieving learning competencies?

Part III. CBAR Title Proposals

After your purposeful observations and critical reflections with your Resource Teacher, think of possible classroom-based action research proposal titles that you intend to pursue in the near future which is aligned to your field of specialization and/or current teaching assignments involving curriculum, instruction/pedagogy (i.e., differentiating learning environments and opportunities to address learner diversity, ensuring learner safety and security), educational technology, school-community engagement, and teacher's personal growth and professional development.

Please come up with at least three titles and three or more research questions in your proposed CBAR titles and fill in the table below. Remember that a good CBAR title should contain TOPIC, ACTION and INTERVENTION.

Proposed CBAR Titles Pertaining to Curriculum, Instruction and /or Pedagogy (including differentiation		
Boosting Afternoon Engagement of Grade 4 Pupils through Radio Calisthenics-Inspired Movement Breaks	**Improving Classroom Waste Management through a Gamified "Clean-Desk" Reward System**	**Increasing Oral Participation Frequency via a Visual Recitation Leaderboard and Token Economy**
Proposed Research Questions	Proposed Research Questions	Proposed Research Questions
hat is the level of student alertness and engagement during the 1–3 PM class period before and after the implementation of movement breaks?	What is the current status of classroom cleanliness and waste disposal habits among the students?	What is the baseline frequency of student recitation during daily Science or Philosophy discussions?
How do students perceive the impact of the "pause-and-dance" routine on their readiness to learn?	How does a competitive reward system influence students' sense of responsibility for their immediate environment?	How does the visible tracking of recitation frequency affect the motivation of both active and passive learners?
Is there a significant difference in student participation rates during afternoon sessions when physical movement is integrated?	To what extent does the gamified intervention reduce the frequency of littering inside the classroom?	Is there a significant increase in the diversity of students participating in class discussions when a token economy is applied?

Note: Please read this before you answer the worksheet.

Sample CBAR Report

The Use of Contextualized Video Instruction in Science 4 Pupils' Academic Achievement

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ABSTRACT

This study presents the use of contextualized video instruction in Science 4 pupils' academic achievement. The growing demand for flexible learning encourages teachers to use videos to seemingly integrate a contextualized learning approach. The research method is a quasi-experimental mono method pretest-posttest of quantitative classroom-based action research. The researchers used convenience sampling to gather the 32 bonafide fourth-grade pupils (n=32) of the Philippine Normal University - Institute of Teaching and Learning. Researchers found that videos are powerful tools to improve academic achievement; a highly positive rectilinear correlation statistically significant from 29.28 ± 5.577 to 38.09 ± 6.827 ($p < .001$); an improvement of 8.81 ± 1.25 from pretest to post-test. These associations may indicate that videos improve academic achievement in Science 4. Understanding these relations may elucidate potential academic patterns and lead to better treatment designs in online learning.

Keywords: contextualized video instruction, contextualized learning approach, academic achievement, flexible learning

1 Introduction

The Local Learning Modalities Offered in Online Learning

One of the three Science domains is for learners to comprehend and apply scientific information in a glocal setting through contextualized learning (K-12 Science curriculum guide, 2016, p.2). With concerns about the rapid transmission of the COVID-19 virus, the Department of Education offers Self-Learning Modules (SLMs) with alternate learning delivery modalities to Filipino students (DepEd, 2020).

In the same article (2020), DepEd Region II Director Dr. Estela Carino said that most activities use contextualized filmed courses taken nationwide. The department would like to review the 1999 Dynamic Learning Program (DLP) of Dr. C. Bernido and Dr. MVC. The DLP encouraged self-directed learning in Science, Technology, Engineering, and Mathematics (STEM).

The Current Problem in an Authentic Online Learning

Unfortunately, many SLMs and video tutorials featured assumptions that hampered educational quality (Adonis, 2021). According to Liza Marie Campoamor Olegario, an educational psychologist at the University of the Philippines College of Education, remote learning implementation is excessively teacher-centered compared to the aims of a 21st-century learner, that is, an authentic student-centered approach (Moradi, 2018). As a result, Olegario asserted that the existing curriculum is overly subject-centered and misaligned with what learners require at their particular ages.

PNU-ITL's Bichronous Learning Modality

The Philippine Normal University-Institute of Teaching and Learning (PNU-ITL) launched its first bichronous mode this Academic Year 2020-2021, bringing national concerns into the classroom environment. One of its teaching strategies is to employ contextualized videos associated with Themes and Sub-themes created by PNU-ITL curricularists and academicians. With these in circulation, there is great relevance to conducting a study on how contextualized video improves Science academic attainment.

1.2 Statement of the Problem

This action research in the classroom evaluated the effectiveness of employing contextualized video instruction as instructional resources to improve fourth-grade pupils' academic achievement. This study sought to answer the following questions:

1. As perceived by fourth-graders, what are the elements of an effective contextualized video instruction for academic achievement?
2. What is the performance of fourth-graders in Science in terms of pre-test and post-test?
3. Are there any significant differences between the pre-test and post-test scores when teachers give contextualized video instructions?
4. Is there a significant relationship between the pre-test and post-test scores when teachers supplied contextualized video instructions?

1.3 Review of Related Literature and Studies

Theoretical definition of Contextualized Video Instruction

Videos are classified based on preparation, presentation techniques, and educational goals (Kostereliogiu, 2016; Kay, 2012). Contextualized videos imply sensory experiences in delivering educational programs for all learners (Kurt, 2017; Kinash & Kung, 2016; Gagne, 2013). Parallel to this study, Shah & Khan (2015) present the use of video instruction as a way to present "phenomena and processes vividly, simulate complex content, and present different levels of abstraction."

Thematic Instruction and Contextualized Video Instruction in PNU-ITL

Brame (2017) listed three elements to create and implement quality video content, namely (a) cognitive load, (b) student engagement, and (c) active learning. A teacher must align the former elements to three aspects of teaching contextualized science units (Dolfing et al., 2020). The first is to create a context in class. Similar to the PNU-ITL procedure of quarterly themes and weekly subthemes, foci are given to students every Monday anchored across all subjects to create a seamless academic journey. Also, teachers transform the learning space from teacher-centered to an authentic approach. Finally, teachers prepare new and seamed content age-appropriate for learners. Such was the case of Laksana et al. (2019) in using ADDIE (analyze, design, develop, implement, evaluate) model to create contextualized thematic electronic learning materials while also considering flipping online learning into chunks specified in Most Essential Skills (Williams & Gils, 2018).

Contextualized Video Instruction in Science-learning

Results have shown that Science-based video instruction is more effective than traditional teaching (Shah & Khan, 2015). Wilujeng et al. (2020) emphasized the tool to improve academic achievement using a pretest-posttest design approving a study on improved concept retention and concept attainment (Kostereliogiu, 2016; Cho & Johnson, 2005). With this in mind, teachers should trigger higher-order thinking skills to create a permanent recording environment that captures the natural phenomena and unlimited case analyses (Tan & Towndrow, 2009; Rasi & Poikela, 2016).

Utilizing videos in learning environments is valuable for independent, concrete, and perpetual learning (Muhammad, 2018). The more senses in the learning environment, the simpler and more significant the lesson will be (Dale, 2018). In video practices, pupils master authentic objects and development sequences by perception, and the recordings give freedom for in-depth learning (Castro-Alonso & Atit, 2019). Therefore, video instruction is a productive type of communication.

Laws in Determining Age-Appropriate Video Content

In creating video content suitable for young audiences, there are considerations that need to be considered before making one. YouTube Help provides the United States Federal Trade Commission (FTC) guidelines (2019) in child-friendly video content. Two conditions must be met to make a child-friendly video, one of which is that children are the main audiences of the intended video. Another is that, video creators must evaluate their videos using the following questions below:

a. Is the video educational?
b. Is it intended for children?
c. Is the language simple to understand?
d. Does it include songs, stories, or poems?
e. Does it include activities (e.g. play-acting, simple songs or games, or early education)?
f. Does it involve characters, celebrities, or toys that appeal to children, including animated characters or cartoons?

Localizing these questions into the Philippine setting, Republic Act No. 8370 or the Children's Television Act of 1997 provides a provision on the "importance and impact of broadcast media, particularly television programs, on the value formation and intellectual development of children." Chapter VIII *The Television Violence Rating Code*, sec. 22. *The Charter of Children's Television* stipulated seven provisions a video creator must strictly adhere to. The following stipulations are directly quoted from the law.

a. "Children should have high quality programs, which are made specifically for them and which do not exploit them. These programs, in addition to being entertaining, should allow children to develop physically, mentally, emotionally, morally, and socially to their fullest potential;
b. Children should hear, see, and express themselves and their culture, languages, and life experiences through television programs, which affirm their sense of self, community, and place;
c. Children's programs should promote an awareness and appreciation of other cultures in parallel with the child's own cultural background;
d. Children's programs should be wide-ranging in genre and content, but should not include gratuitous scenes of violence and sex;
e. Children's programs should be aired during regular time slots when children are available to view and distributed through widely accessible media or technologies;
f. Sufficient funds must be made available to make these programs conform to the highest possible standards; and
g. Government, production, distribution, and funding organizations should recognize both the importance and vulnerability of indigenous children's television and the steps to support and protect it.

1.4 Theoretical Framework

This study anchors on Edgard Dale's Cone of Experiences.

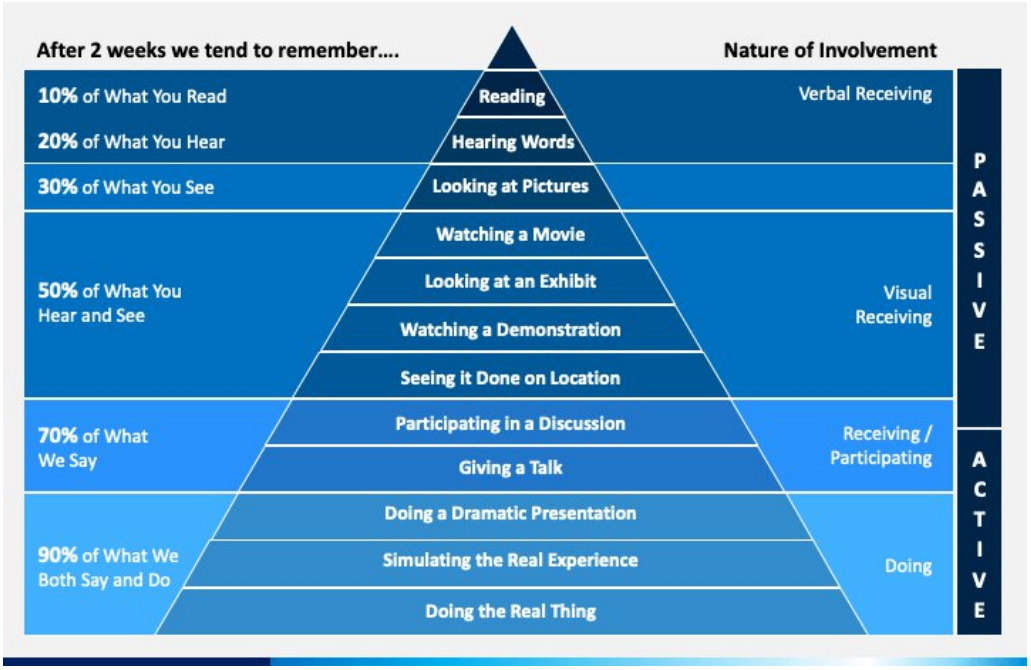

Fig. 1. Edgard Dale's Cone of Experiences

A graphic depiction of the sorts of experiences that people have as they prepare to synthesize knowledge and transform it to understanding. Dale said that humans may have a variety of experiences. Because knowledge is a process, each degree of experience results in a different level of efficacy.

As he moves down from top to bottom, he divides his model into three broad categories, from most abstract to least abstract. Dale claims that the arrangement in the cone is based on abstraction and the number of senses engaged. The experiences at each level can be combined and interconnected, resulting in more meaningful learning.

Practicing with items is an example of inactive or direct experience (the student actually ties a knot to learn knot-tying). This type of experience entails tangible, immediate action as well as the use of one's senses and body.

Interpreting photos and drawings is part of the iconic experience (the student looks at drawings, pictures or films to learn to tie knots). Such encounters are once separated from the physical reality and restricted to two or three senses.

Reading or hearing symbols (the learner reads or hears the word "tie" and develops a picture in his or her mind) is part of the symbolic experience. In such cases, the action is usually indifferent, and the experience is restricted to thoughts and ideas.

Elements of a Video Instruction

The power of video instruction has led many pupils to link concrete events and phenomena to scientific principles, vice versa. Although practice teachers and teachers alike use low-cost cameras and free editing software, teacher-made video targets higher-level thinking skills with remedial tools like guide questions and supplementary activities catering to their needs and interests (Bates, 2016).

The words and images teachers use for teaching have a significant and lasting influence. An animation that is indirectly related to the content might hamper rather than enhance learning. Pupils access learning materials via online classrooms or websites like Google Classroom and Youtube. The varying quality of pupils' electronic devices creates inequity in content dissemination. While the video is a powerful tool, this should have keywords for learners with hearing disabilities. Teachers should meticulously select supplementary activities for pupils who reject videos that require analysis or interpretation (Bates, 2016).

Table 1 below simplifies the list of recommendations of Guo et al. (2014), Thomsen et al. (2014), Ibrahim et al. (2012), deKoning et al. (2009), Zhang et al. (2006), Mayer & Moreno (2003) succeeding statements suggest the following actions that teachers may use based on sound research in creating video content, namely signaling, segmenting, weeding, and matching modality.

Table 1. Effects and examples of the four processes in considering video elements.

PROCESSES	EFFECT/S ON COGNITIVE LOAD	EXAMPLES
Signaling: highlighting important information	Reduce extraneous load Enhance germane load	● key words ● color or contrast to associate information ● learning objectives
Segmenting: summarizing information	Manages intrinsic load	● guide questions ● length of videos (3-6

	Enhance germane load	minutes)
Weeding: removing unnecessary information	Reduce extraneous load	● music ● complex backgrounds
Matching modality: using visual and auditory channels to show information	Enhance germane load	● Khan-style tutorial videos ● narrated animations

Teacher's Role in Asynchronous Modality and Other Considerations

According to Sosnowski (2020), the role of the teacher as a facilitator allows pupils to complete asynchronous tasks at their own pace given that time is flexible in the New Normal; however, classroom management becomes an arduous task. Some pupils may also spend more time accessing the materials than focusing on its content. Finally, when a pupil has a weak internet connection, video instructions are slow to load due to the large file size. There are video streaming platforms that may help this issue, yet a decent internet connection will run these platforms.

These learning materials may retain scientific concepts or drawbacks internally (learning objectives, content, video production) or externally (bandwidth, lack of gadgets, poor internet connection). Mixed learning offers instructive solutions that are video-based learning offered close by other pedagogical tasks and techniques, offered both on the web and face to face, can make for a very beneficial learning experience and solid outcomes (Kinash & Kung, 2016).

Synthesis

This review of related literature and studies allows practice teachers to design contextualized videos aligned with the Cognitive Theory of Multimedia Learning to sensitize academic achievement with contextualized content glocally. Specifically, the effectiveness of using contextualized video instruction in Science 4 shall depend on its validity and reliability on existing and credible published scientific and academic journals parallel with this research; thus, these studies will strengthen the acclaimed results and conclusion.

1.5 Conceptual Framework

Based on the theoretical discussion, the researchers present the conceptual framework of the study.

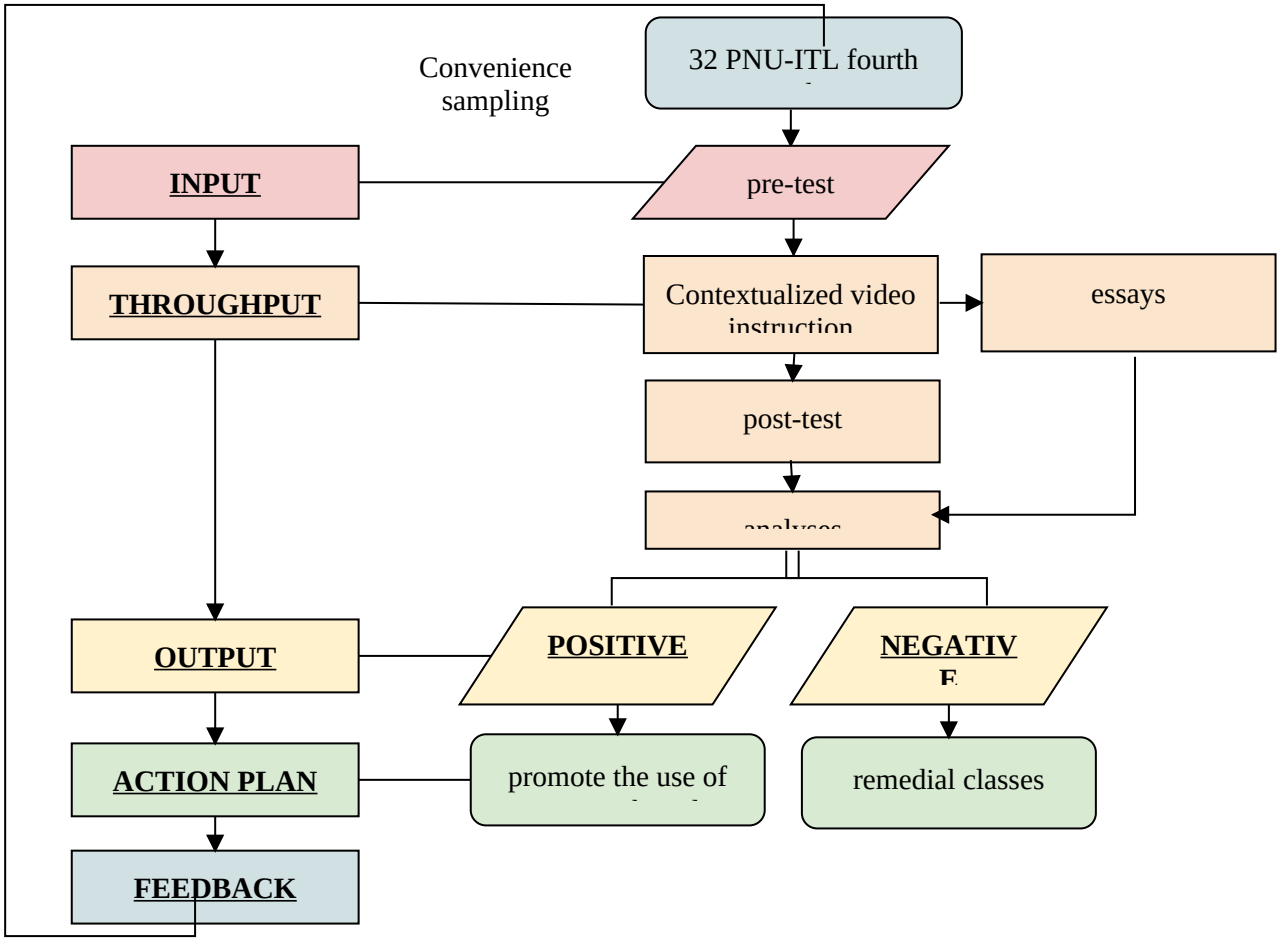

Fig. 2. The research process archetype

Figure 2 depicts the five stages organized according to how the researchers performed the investigation. The first stage, input, is the administration of the pre-test. The pre-test is a 50-item objective type test with four alternatives in a multiple-choice style (A, B, C, D). The themes chosen are in line with both the 2016 K-12 Science Curriculum Guide and the PNU-ITL Interdisciplinary Curriculum Plan. The researchers administered the pre-test before the commencement of the Fourth Quarter.

Following the pre-test administration, the researchers hold synchronous sessions lasting no more than an hour to enhance subject learning. With the screen time health protocols, the contextualized video instruction (independent variable) served as additional tools for students to view. In addition, the researchers included essays to measure pupils' ability to relate science concepts authentically.

After a week, the researchers administered a 50-item post-test. The researchers opted to divide the 50-item multiple-choice objective type examination into five subtests instead of a one-time test. A subtest has ten items. As a result, the post-test took five weeks to complete. These were administered on Google Forms and Google Meet, much as the pre-test.

Following the collection of all data, the researchers undertook quantitative analysis. The researchers determined whether to pursue the use of contextualized videos or perform remedial sessions to remediate and enforce understanding of the lesson (dependent variables).

If the fourth-graders' academic achievement increased, then the researchers may promote the use of contextualized videos for the succeeding academic year. On the contrary, future action may entail conducting remedial classes for those pupils who have not satisfied the competencies.

The researchers assumed that the system is deterministic or stochastic, in which case the pre-test and outcomes were predictable. To synthesize, the researchers relied on empirical data.

1.6 Limitations and Delimitations

With a tantamount body of knowledge accessible in the online platform, there is a possibility that some research results have biased perspectives in which practice teachers would conduct the study to reassure existing claims. More so, these researches were in the pre-pandemic times. Research results were for a face-to-face mode of learning. In that case, there is a limited amount of research and journals related to using contextualized videos in online learning. Also, most literature and studies focus on high school learners and college undergraduate students. This academic level limits practice teachers to assess the use of contextualized video instruction at the elementary level. The use of contextualized video instruction in online learning lacks; hence, research results may change.

As the researchers are elementary practice teachers, they delimit this study to 32 fourth-grade pupils of PNU-ITL using convenience sampling. It was necessary because the researchers are the practice teachers of the said sample.

1.7 Definition of Terms

Below is the list of the definitions of terms that transpired in this study. The terms with asterisk were defined operationally, while those without were defined conceptually.

a. ***Academic achievement** -** Academic achievement refers to the fourth-graders' Science overall test scores in the pre-test and post-test.
b. ***Bichronous learning modality** -** A bichronous learning modality is a form of online learning modality implemented by the Philippine Normal University-Institute of Teaching and Learning this Academic Year 2020-2021. This divides learning proper into two: synchronous and asynchronous sessions, where synchronous sessions are done via Zoom within an hour, while asynchronous sessions are done individually respective of the tasks given to pupils.
c. ***Contextualized learning approach* -** A contextualized learning approach is one type of learning approach that deals with concepts relating to meaningful situations relevant to pupils' lives to help them effectively.
d. ***Contextualized video instruction** -** A contextualized video instruction is a form of multimedia presentation that contains pictures, sounds, learning objectives, and learning content. This refers to YouTube videos, and/or teacher-made videos used in asynchronous sessions to facilitate context based on PNU-ITL's sub themes per week.
e. ***Cognitive load* -** A cognitive load is the amount of effort required while reasoning and thinking
f. ***Extraneous load* -** An extraneous load is a component of the cognitive load that may be unnecessary toward the desired learning outcome (irritating music, extra animations).
g. ***Flexible learning* -** Flexible learning is a form of learning intervention that responds to the needs, context, circumstances, and diversity of learners.
h. ***Germane load* -** A germane load is a component of the cognitive load that features higher order thinking skills (analyzing, evaluating, creating) to master the desired learning outcome.
i. ***Intrinsic load* -** An intrinsic load is a component of the cognitive load that features connective

subjects under study (word pairs/vocabularies, and grammar).

j. ***Self-directed learning*** **-** Self-directed learning is a form of learning intervention that allows learners to learn the subject content on their own by actively using the learning materials given. This may not entail maximum supervision of the teacher.

2 Methods

2.1 Research Design

The research design is a quasi-experimental mono method pretest-posttest classroom-based action research. A quasi-experimental study is an interventional study that employs statistical design principles to estimate the impact of a given intervention on a specific population (Dinardo, 2008). The researchers used quantitative methods in assessing pupils' academic achievement through objective-type pre-test and post-test, and essays. Three Science instructors validated the objective-type tests. To classify pupils' level of connecting scientific concepts into the subthemes, the researchers adapted the International Reading Association NCTE Essay Writing Rubric (2013) to numerically grade the essays. The researchers examined academic achievement to enhance learning resources through classroom-based action research (Mertler, 2012).

2.2 Research Locale

The Institute of Teaching and Learning (ITL) is a laboratory school of the Philippine Normal University. It was established in 1905 and was originally the Philippine Normal Training Department headed by Helen Blue. Over time, PNU-ITL caters to the needs of elementary and secondary learners. The institute is also a training center for fourth-year undergraduate practice teachers of the different faculties and institutes of the university.

ITL is implementing a bichronous mode of online distance learning. An hour is allocated for synchronous sessions, while asynchronous sessions are for paperwork and assignments. With the flexible scheme, ITL curricularists and professors implemented the Interdisciplinary Curriculum Plan that allows malleability to the topics horizontally and vertically aligned to a quarter's theme. Each week of a quarter has a sub-theme that tackles a more holistic and sensible approach to the learners' needs and interests. The institute expects teachers and practice teachers to integrate their lessons on the prescribed subthemes while maintaining the integrity of the lesson competencies handed down by DepEd. Specifically, the fourth quarter's theme is My World. Table 2 itemized the following sub-themes, focus points, and questions below.

Table 2. PNU-ITL's fourth quarter theme data

WEEK	SUBTHEME	ENDURING UNDERSTANDING	ESSENTIAL QUESTION
1	Characteristics	The earth is a unique planet with certain characteristics (e.g. only planet with life); movement, mechanisms	How will you describe the different characteristics of the earth?
2	Culture	The world is made up of	How do I respect the different

		diverse people with different traditions and cultures.	cultures of the world?
3	Worldviews	The people of the world possess different beliefs and values as influenced by their worldviews.	Why do the people of the world have different cultures and values?
4	Science, Technology, Society	Science and technology play an important role in society.	How are science and technology important to society?
5	Global Understanding & Call for Action	As the people of the world are diverse, conflict may arise yet we should learn to co-exist to have peace and security.	How to live with peace and security amidst conflicts?

2.3 Respondents of the Study

The respondents were the 32 grade four bonafide pupils of Philippine Normal University - Institute of Teaching and Learning (PNU-ITL). The respondents were assumed to be credible, neurotypical, and uphold academic honesty. A “typical” grade four pupil is a 10-year-old (f=19, 59.4%) female (f=18, 56.3%) who lives in the City of Manila (23, 71.9%) who has one to two siblings (f=22, 68.8%). She is the youngest (f=13, 40.6%) among her siblings and of Tagalog descent (f=24, 75.0%), whose religion is Roman Catholic (f=20, 62.5%), and a bonafide pupil of PNU-ITL (f=25, 78.1%). She also uses a laptop (f=25, 78.1%), smartphone (f=11, 34.4%), and tablet (f=10, 18.5%).

2.4 Instrumentation

The instruments used were Learner’s Profile, 50-item pre-test and post-test (table of specifications), essay, and a rubric.

Learner’s Profile

The Learner’s Profile (LP) enabled researchers to learn about the demographics of grade four respondents. The researchers adopted Duad's (2016) “Development of Strategic Intervention Material (SIM) for grade 10 Science: Chemical equations and reactions [Manuscript submitted for publication]. It consists of the respondents’ name, grade and section, age, residence, sex, number of siblings, birth order, ethnic group, religion, and pre-school graduated from. The LP was transferred from pen-and-paper format to Google Forms.

50-item pre-test and post-test

The researchers created the 50-item pre-test and post-test. Three Science teachers validated the said teacher-made tests. The coverage was aligned with the competencies of the 2016 Science Curriculum

for grade 4. The tests are in multiple-choice objective type format. Each question consists of four alternatives (A, B, C, D), arranged alphabetically and/or by length. The tests range from Remembering to Evaluate of the Revised Bloom's Cognitive Taxonomy.

5-item Essay

The 5-item essay is a narrative of pupils' understanding of the relationship between science concepts to the sub themes promoted by PNU-ITL. After pupils watched an asynchronous video, pupils answered a guide question in the form of an essay. The researchers used Google Forms to prevent plagiarism or copying of answers. They also used a third-party plagiarism scanner like Duplichecker and Small SEO tools and Grammarly.

Rubric

The rubric was adapted from the International Reading Association (2013). It originally contained five criteria, specifically (1) focus and details (FD), (2) organization (OR), (3) voice (VO), (4) word choice (WC), and (5) sentence structure, grammar, mechanics, and spelling (SGMS). However, the researchers opted to delete WC as it is similar to OR and VO. The rubric is structured holistically with a rate from 1 as the lowest to 4 as the highest. These criteria were averaged to arrive at the final score.

2.5 Data Gathering Procedure

Practice teachers conceptualized using contextualized video instruction last Third Quarter of the Academic Year 2020-2021 (January to March 2021), but their implementation started in the first week of the Fourth Quarter (April 26, 2021). It ran until the fifth week of the Fourth Quarter (June 3, 2021). Pupils used Google Forms to answer the tests. After conducting the tests, the researchers performed analyses using parametric tests in SPSS. The statistical tests used are in the succeeding paragraphs.

2.6 Statistical Treatment of Data

The researchers tabulated the responses electronically to facilitate statistical analyses. They used the methods of estimating reliability to determine the reliability of the tests. Some of which are stipulated as follows:

1. stability: test-retest method using Pearson r;
2. internal consistency: Kuder-Richardson 20 and Cronbach's alpha

On the other hand, they ran the data using the following statistical tests: percentage, frequency distribution, sum, arithmetic mean, standard deviation, coefficient of variation, and paired t-test. The succeeding tables show the range of interpretations of some statistical tests.

Table 3. Interpretation of Pearson correlation coefficient for test-retest method

RANGE	REMARK
$\lvert r \rvert = 1.000$	Perfect positive/negative rectilinear correlation

$0.801 \leq \lvert r \rvert \leq 0.901$	Very high positive/negative rectilinear correlation
$0.701 \leq \lvert r \rvert \leq 0.800$	High positive/negative rectilinear correlation
$0.601 \leq \lvert r \rvert \leq 0.700$	Marked substantial positive/negative rectilinear correlation
$0.401 \leq \lvert r \rvert \leq 0.600$	Moderate positive/negative rectilinear correlation
$0.201 \leq \lvert r \rvert \leq 0.400$	Low positive/negative rectilinear correlation
$0.001 \leq \lvert r \rvert \leq 0.200$	Negligible positive/negative rectilinear correlation
$\lvert r \rvert = 0.000$	No rectilinear correlation

Table 4. Interpretation of Cronbach's alpha

RANGE	REMARK
$\alpha \geq 0.91$	Excellent
$0.81 \leq \alpha < 0.90$	Good
$0.71 \leq \alpha < 0.80$	Acceptable
$0.61 \leq \alpha < 0.70$	Questionable
$0.51 \leq \alpha < 0.60$	Poor
$\alpha < 0.50$	Unacceptable

2.7 Ethical Considerations

This classroom-based action research is in full compliance with research ethics norms concerning the Republic Act 10173 or the Data Privacy Act of 2012 of the National Privacy Commission. The data collection involves human respondents of typical grade four pupils of age 10. Furthermore, this required a synchronous interaction of the researchers and the respondents within an hour for pre-test and post-test. The core responsibility of the researchers is to explain in detail the research to the respondents. Given the time constraints of an hour of synchronous sessions in a week, pupils took the post-test every Thursday.

The research respondents signed the agreement that outlines the purpose, the research, and the form of dissemination and usage in a format in Google Forms. The agreement consists of the respondents' wish to retract information or withdraw participation at any point in the test time, the anonymity, and the confidentiality of their data. Their participation was voluntary.

After the data collection, all data was systematized and stored in one password-protected external storage drive. The principal researcher secured the storage drive. All information was kept confidential. The research team and panelists will only access the data only for cross-references.

3 Results and Discussion

The discussion follows Chapter 1 Introduction order of problems. The researchers follow the standard format: graphs/tables, then textual descriptions.

3.1. As perceived by fourth-graders, what are the elements of an effective contextualized video instruction for academic achievement?

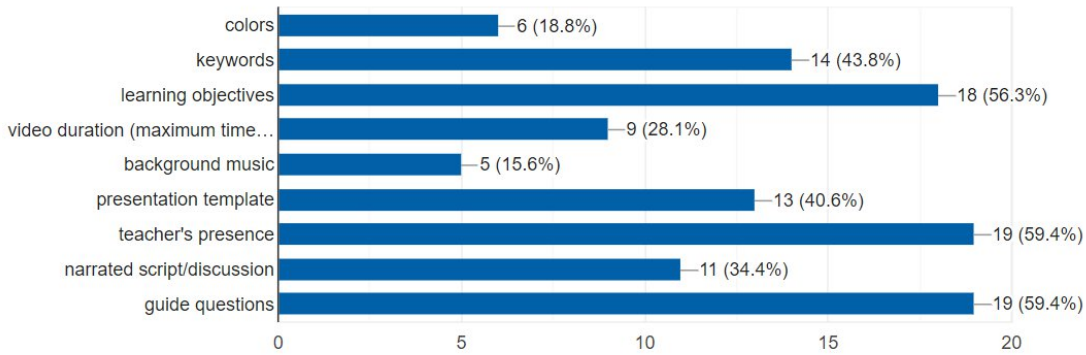

Fig. 3. Graphical representation of respondents' preferred elements in a video (n=114)

Based on figure 3, fourth-graders topped the options *teacher's presence* and *guide questions* with 19 (59.4%) responses. Followed by *learning objectives* with 18 (56.3%) responses and *a presentation template* with 13 (40.6%) responses. Only a few responses considered *colors* (f=6, 18.8%), and *background music* (f=5, 15.6%) as elements of an effective contextualized video instruction. These data are essential in creating teacher-made videos as preferred by fourth-grade pupils.

Similar to the recommendations of Guo et al (2014), Thomsen et al. (2014), Ibrahim et al (2012), deKoning et al (2009), Zhang et al. (2006), Mayer & Moreno (2003), the fourth-graders prioritized the matching modality and segmenting processes. Followed by the signaling process in relation to colors. Finally, less responses were included in the weeding process.

This result may imply that fourth-graders tend to look at the manner in which videos are created more than the creative features of the video. Specifically, the matching modality process shows how information is being conveyed. Consequently, this may imply that fourth-graders considered the time they may take to watch the videos as well as guide questions included. The guide questions are there to allow pupils to answer a question after watching the video.

3.2. What is the performance of fourth-graders in Science in terms of pre-test and post-test?

Table 5. Normality results of the scores on 50-item pre-test and post-test using Shapiro-Wilk test

	df	stat.	p

pre-test	32	.970	.505
post-test	32	.971	.529

A Shapiro-Wilk (S-W) test determined the normality distribution of the scores of the 50-item tests. Normality tests are vital to examine the data in parametric testing. Since the S-W test is more appropriate for sample sizes with less than 50 respondents, in this case, 32 respondents, then S-W is fit to use.

The 50-item pre-test follows the normal distribution, W(32)=0.970, p=0.51. Similarly, the 50-item post-test normality results suggest the scores' normal distribution with W(32)=0.971, p=0.529. Both pre-test and post-test probability (p) are greater than .05; therefore, the researchers retain the null hypothesis of population normality.

With the test scores marked as normal, the researchers used parametric tests to analyze succeeding data.

Table 6. Statistical results of pre-test and post-reliability's measure of internal consistency using KR-20

	Cronbach alpha	N of items	M	SD	CV
pre-test	.705	50	29.28	5.577	.191
post-test	.855	50	38.09	6.827	.179

* based on M

A Kuder-Richardson 20 (KR-20) assessed the internal consistency reliability of both pre-test and post-test. Statisticians used KR-20 for dichotomous responses (Yes/No; Correct/Incorrect). The number of items indicates those identified to have variances. In this context, the researchers evaluated the results across items within the tests.

The 50-item pre-test has a Cronbach alpha of .705. Based on the ruling presented in table 4, the pretest internal consistency is acceptable. Besides, the post-test has an alpha of .855, which is interpreted as good. In general, as the Cronbach alpha increases, the inter-correlations among test items also increase (Hajjar, 2018).

Take note that the internal consistency reliability test only provides a measure that each item is measured correctly and reliably and may show how pupils' response correlate with one another and not their acquired scores per se, yet this may show us a background of how the test items are reliable indicators to measure the prescribed learning competencies.

Table 7. Descriptive data on acquired scores for pre-test (n = 32)

Level	Range	f	%	M	SD	CV
Superior	41 - 50	0	.0	.00	.000	.000

Advance	31 - 40	15	46.9	33.87	2.850	.084
Proficient	21 - 30	14	43.7	26.57	2.953	.111
Approaching Proficiency	11 - 20	3	9.3	19.00	1.000	.053
Beginning	00 - 10	0	.0	.00	.000	.000
OVERALL		32	100	15.89	1.361	.049

* based on M

In the 50-item pre-test, 46.9% of pupils were advanced (M = 33.87, ± 2.850); followed by the 43.7% of the class who were proficient (M = 26.57, ± 2.953). Finally, only 9.3% were approaching proficiency (M = 19.00, ± 1.000; CV = .053). The results may imply that fourth-grade pupils are adept learners before the conduct of fourth quarter.

Table 8. Acquired scores on a 5-item essay (n = 32)

Level	Range	f	%	M	SD	CV
Superior	16.1 - 20.0	18	56.3	19.31	.864	.045
Advance	12.1 - 16.0	6	18.7	14.87	1.339	.090
Proficient	8.1 - 12.0	4	12.5	11.19	.898	.080
Approaching Proficiency	4.1 - 8.0	3	9.3	7.50	.500	.067
Beginning	0.0 - 4.0	1	3.1	2.50	----	.000
OVERALL		32	100.0	11.07	.720	.056

* based on M

Table 8 describes the acquired scores on a 5-item essay that pupils answered after watching contextualized video instruction. The researchers ensured that all essay answers went under careful content analysis and plagiarism detection using third-party softwares like Grammarly, Small SEO tools-online plagiarism detector, and Duplichecker. 56.3% of the class were superior in giving narratives in relation to science topics (M = 19.31, ± .864). 6% were advanced (M = 14.87, ± 1.339) and only one pupil is under Beginning level.

Results have shown that video teaching explains phenomena and processes, replicates complex information, and conveys diverse abstract ideas more than traditional teaching (Shah & Khan, 2015). Such was the case when grade 4 pupils answered the guide questions in Google Form. Concepts transpired in essays authorizing researchers to assess concept retention and concept attainment in authentic problems (Wilujeng et al. 2020; Castro-Alonso & Atit, 2019; Kostereliogiu, 2016; Cho & Johnson, 2005). With essays as supplementary data of this study, pupils capture scientific concepts in real-life settings (Tan & Towndrow, 2009; Rasi & Poikela, 2016).

The majority of the scores above proficiency may imply that pupils could relate scientific concepts into real-life situations with the use of contextualized video instruction. More so, this may suggest that video information is assimilated according to the pupils' prior knowledge of the topics.

Table 9. Acquired scores on a 50-item post-test (n = 32)

Level	Range	f	%	M	SD	CV
Superior	41 - 50	12	37.5	44.91	2.850	.063
Advance	31 - 40	15	46.9	36.20	2.953	.081
Proficient	21 - 30	5	15.6	27.40	1.000	.036
Approaching Proficiency	11 - 20	0	.0	.00	.000	.000
Beginning	00 - 10	0	.0	.00	.000	.000
OVERALL		32	100	21.70	1.361	.036

* based on M

After an extensive process, table 8 shows that 37.5% of the class excelled and were superior after watching the video instruction (M = 44.91, ± 2.850). 46.9% of the class were advanced (M = 36.20 ± 2.953). Finally, 15.6% of the class were proficient (M = 27.40, ± 1.000). None fell in the Approaching Proficiency and Beginning levels.

Similar to the pre-test and post-test's internal consistency in the previous tables, fourth-graders improve based on their score levels. If we were to simply compare the frequency of pupils based on their academic achievement level between pre-test and post-test, pupils improved in post-test with 12 pupils in the superior level and none fell below the proficient level. Consequently, if we were to compare their means, the post-test scores have greater means compared to the pre-test. On the other hand, the standard deviation for both scores are the same even though the pupils' level changed. Hence, comparing the coefficient of variation is necessary.

One feature of coefficient of variation is that the lower the ratio of the standard deviation to mean is, the lesser its variability and high in stability. In this study, it manifests that post-test scores have lesser variability compared to pre-test scores.

3.3. Are there any significant differences between the pre-test and post-test scores when teachers give contextualized video instructions?

Table 10. Correlation between pre-test and post-test scores using Pearson r (n = 32)

		pre-test	post-test
pre-test	Pearson r	1	.707
	Sig. (1-tailed)		.000

post-test	Pearson r	.707	1
	Sig. (1-tailed)	.000	

** Correlation is significant at the 0.01 level (1-tailed).

To determine whether the stability between pre-test scores and post-test scores is stable or not, researchers assess the r-value towards 1 or -1. The stronger the association between pre-test scores and post-test scores, the more stable the scores are. The researchers used Pearson r to analyze the stability between pre-test scores and post-test scores. There is a highly positive rectilinear correlation between the scores of the two tests, which was statistically significant ($r = .707$, $n = 32$, $p = .000$).

3.4. Is there a significant relationship between the pre-test and post-test scores when teachers supplied contextualized video instructions?

Table 11. Paired t-test result of pre-test and post-test scores

	M	**SD**	**CV**	$t_{critical}$	t_{value}	**df**	**Sig.**$_{2\text{-tailed}}$
post-test	38.09	6.827	.036				
pre-test	29.28	5.577	.049				
OVERALL	8.81	1.250	0.141	2.04	10.197	31	.000

Finally, the pre-test and post-test scores went under a paired t-test to determine whether there was a statistically significant mean difference between the post-test scores when pupils watched contextualized videos compared to the pre-test scores. Based on table 11, $t (31) = 10.197$, $p < .001$.

Based on all analyses, the researchers suggest that there is a highly positive rectilinear correlation statistically significant improvement in academic performance in Science 4 following the use of contextualized video instruction as a tool for online asynchronous learning from 29.28 ± 5.577 to 38.09 ± 6.827 ($p < .001$); an improvement of 8.81 ± 1.250.

All results of this study are parallel to Wilujeng et al. (2020), Castro-Alonso & Atit (2019), Tan & Towndrow (2009), Williams & Gils (2018), Kosterelioglu (2016), Rasi & Poikela (2016), Shah & Khan (2015), and Cho & Johnson (2005). These researchers supported this study that contextualized video instruction in Science learning helps pupils to improve academic achievement. As Wilujeng et al. (2020) used the same design to assess the same variables, this study equally supports a recent claim of a tool in online learning. With that said, contextualized video instruction captures unlimited case analyses that pupils may answer parallel to Tan & Towndrow (2009) and Rasi & Poikela (2016). Consequently, the process of answering questions in a self-paced manner increased pupils' academic achievement creating an in-depth learning (Castro-Alonso & Atit, 2019; Muhammad, 2018).

4 Conclusions and Recommendations

The usefulness of adopting contextualized video training as a technique to improve academic attainment was investigated in this study. With the new learning scheme in place, it is critical to evaluate the success of using videos that are contextualized globally. The researchers created videos integrated on the themes and subthemes of the Philippine Normal University - Institute of Teaching and Learning (PNU-ITL) Interdisciplinary Curriculum Plan.

Following a thorough examination of the data, the researchers determined that contextualized video training should have the teacher's presence, guide questions, and learning objectives. Based on the improvements in post-test scores, the researchers therefore conclude that contextualized video instruction can be used as a tool for asynchronous learning.

The researchers encourage PNU-ITL faculty and practice teachers to continue to employ contextualized video instructional resources tailored to the requirements and interests of students. A needs assessment might be carried out at the start of the academic year. Unfortunately, unlike a feature in Moodle ePNU, this study was unable to assess essays using a conventional evaluation instrument easily as the researchers used third-party applications.

Future researchers may assess pupils' former quarterly test grades concerning their study. They may also conduct a test with a control group that does not use contextualized video instruction and an experimental group that uses the said tool. The researchers may also suggest assessing how contextualized video instruction affects pupils' academic achievement in terms of the Cognitive Process Domain. These may change the design of the study and scope for a better analysis.

Besides using essays as an assessment tool, educators may vary their assessment to evaluate other domain processes, such may include pupils' assignments in data analysis. With issues on health protocols and continuity of online learning, it is expected that teachers strategize chunking learning in asynchronous mode given time flexibility.

The authors declare no conflict of interest in the preparation of this paper. However, due to test constraints in the online administration of pretest and posttest, Google Forms limits the authors to detect possible cheating and/or plagiarizing; hence, succeeding authors should use different test modalities that will ensure cheat- and/or plagiarize-proof data.

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APPENDICES

APPENDIX A: LEARNERS' PROFILE

		f	%
AGE	9	13	40.6
	10	19	59.4
	Total	32	100.0
RESIDENCE	City of Manila	23	71.9
	Outside City of Manila but within Metro Manila	5	15.6
	Outside Metro Manila	4	12.5
	Total	32	100.0
SEX	Male	14	43.7
	Female	18	56.3
	Total	32	100.0
NUMBER OF SIBLINGS	none	5	15.6
	1-2	22	68.8
	3-4	4	12.5

	5 or more	1	3.1
	Total	32	100.0
BIRTH ORDER	Only child	5	15.6
	eldest	10	31.3
	middle	4	12.5
	youngest	13	40.6
	Total	32	100.0
ETHNIC GROUP	Tagalog	24	75.0
	Others (Aklanon, Batangueno, Bicolano, Caviteno, Cebuano, Ilokano, Tausug, Visaya)	8	25.0
	Total	32	100.0
RELIGION	Roman Catholic	20	62.5
	Non-Catholic Christian	10	31.2
	Islam	2	6.3
	Total	32	100.0
PRE-SCHOOL	Originally from PNU-ITL	25	78.1
	Private sectarian	4	12.5
	Private nonsectarian	2	6.3
	Public sectarian	1	3.1
	Total	32	100.0

APPENDIX B: VALIDATED TEACHER-MADE TEST EVALUATION

Listed below are facets that evaluators assessed using a forced-choice scale, where 4 as the highest and 1 as the lowest. The following remarks are based on the range of acceptability presented in Table B.0.

Table B.0. Acceptability range based on Mean

RANGE	REMARK

$3.01 \leq M < 4.00$	Highly Acceptable (HA)
$2.01 \leq M < 3.00$	Acceptable (A)
$1.01 \leq M < 2.00$	Questionable (Q)
$0.00 \leq M < 1.00$	Unacceptable (U)

Table B.1. Expert's validation on format and design

FACETS	E1	E2	E3	ACCEPTABILITY		
				M	SD	Remark
The layout is attractive and uniform.	3	4	3	3.33	0.5774	HA
The fonts used are comprehensible and appropriate.	4	4	4	4.00	0.0000	HA
The spacing on the pages contributes to the ease of reading and responding.	4	4	4	4.00	0.0000	HA
Domain 1	3.67	4.00	3.67	3.78	0.1925	HA

Table B.2. Expert's validation on language and style

FACETS	E1	E2	E3	ACCEPTABILITY		
				M	SD	Remark
The language and style is appropriate.	3	4	4	3.67	0.5774	HA
The language used is consistent.	3	4	4	3.67	0.5774	HA
The test is free from grammatical errors.	3	3	3	3.00	0.0000	A
The sentences are properly constructed and punctuated.	3	4	4	3.67	0.5774	HA
The words used in the test are simple and easy to understand.	3	3	4	3.33	0.5774	HA

Domain 2	3.00	3.60	3.80	3.47	0.4619	HA

Table B.3. Expert's validation on content

FACETS	E1	E2	E3	ACCEPTABILITY		
				M	SD	Remark
The items are factual and objective.	3	4	4	3.67	0.5774	HA
The items are free from ideological, sexual, and racial bias.	4	4	4	4.00	0.0000	HA
The items reflect typical scientific bases and/or real-life applications.	3	4	4	3.67	0.5774	HA
The choices for each item are independent and do not overlap.	3	3	4	3.33	0.5774	HA
The items are sound, convenient, useful, and proper for Science 4.	4	4	4	4.00	0.0000	HA
Domain 3	3.40	3.80	4.00	3.73	0.3464	HA

Table B.4. Expert's validation on test specifications

FACETS	E1	E2	E3	ACCEPTABILITY		
				M	SD	Remark
Do the test specifications describe the nature and limits of the achievement domain to be measured?	4	4	4	4.00	0.0000	HA
Are the intended learning outcomes stated in performance measurable items?	4	4	4	4.00	0.0000	HA
Do the test specifications indicate the nature and distribution of items to be included in the test?	4	3	4	3.67	0.5774	HA

Domain 4	4.00	3.67	4.00	3.89	0.1925	HA

Table B.5. Expert's validation on test format and directions

FACETS	E1	E2	E3	ACCEPTABILITY		
				M	SD	Remark
Items arrangement contributes to ease of reading and responding.	4	4	4	4.00	0.0000	HA
Items of the same type are placed together in the test.	3	4	3	3.33	0.5774	HA
Items are arranged in order of increasing difficulty within each section of the test.	4	4	3	3.67	0.5774	HA
The method of answering and the location of the answer space contribute to efficient test taking.	4	4	4	4.00	0.0000	HA
Items are numbered in consecutive order throughout the test.	4	4	4	4.00	0.0000	HA

There is a balanced proportion of correct answers.	3	4	4	3.67	0.5774	HA
The sequence of answers is free from patterns.	3	4	4	3.67	0.5774	HA
Directions provided for the test, as a whole and for separate sections, are clear and concise.	4	4	4	4.00	0.0000	HA
Final copy of the test is satisfactory in terms of spacing, legibility, and freedom from errors.	3	3	4	3.33	0.5774	HA
Domain 5	3.56	3.89	3.78	3.74	0.3208	HA

Table B.6. Expert's validation on test items in general

FACETS	E1	E2	E3	ACCEPTABILITY		
				M	SD	Remark
Each test item is relevant to an important learning outcome.	4	4	4	4.00	0.0000	HA
Each item type is appropriate for the particular learning outcome to be mastered.	4	4	4	4.00	0.0000	HA
Each item presents a clearly formulated task.	4	4	4	4.00	0.0000	HA
Each item is stated in simple, clear language, and free of non functional material.	3	4	4	3.67	0.5774	HA
Each item is free of extraneous clues.	3	3	3	3.00	0.0000	A
The difficulty of each item is appropriate for the task and for the use to be made of the results.	4	4	4	4.00	0.0000	HA
The answer to each item is one that would be agreed upon by experts.	3	4	4	3.67	0.5774	HA
There is an adequate number of items for each interpretation to be made.	4	4	4	4.00	0.0000	HA
The items are sufficiently independent that one item does not aid in answering another.	4	3	4	3.67	0.5774	HA
The items are free from ethnic, racial, ideological, religious, and sexual bias.	4	4	4	4.00	0.0000	HA
Domain 6	3.70	3.80	3.90	3.80	0.1732	HA

Table B.7. Expert's validation on multiple choice items

FACETS	E1	E2	E3	ACCEPTABILITY		
				M	SD	Remark
The stem of each item presents a single clearly formulated problem.	4	4	4	4.00	0.0000	HA
The stem of each item is stated in simple, readable language.	4	4	4	4.00	0.0000	HA
The stem of each item is worded so that there is no repetition of material in the alternatives.	3	4	4	3.67	0.5774	HA
The stem of each item is stated in positive form, whenever possible.	4	4	4	4.00	0.0000	HA
If negative wording is used in the stem, it is emphasized by capitalization, italics, or underlining.	4	4	4	4.00	0.0000	HA
The intended answer is correct or clearly best.	3	4	4	3.67	0.5774	HA
All alternatives are grammatically consistent with the stem and parallel in form.	4	4	4	4.00	0.0000	HA

The alternatives are free from verbal clues to the correct answer.	3	3	3	3.00	0.0000	A
The distractors are plausible and attractive to the uninformed.	3	4	3	3.33	0.5774	HA
The relative length of the correct answer varies to eliminate length as a clue.	3	4	4	3.67	0.5774	HA
Domain 7	3.50	3.90	3.80	3.73	0.2309	HA

Table B.8. Expert's validation on the teacher-made test

DOMAIN	E1	E2	E3	ACCEPTABILITY		
				M	SD	Remark
I. Format and Design	3.67	4.00	3.67	3.78	0.1925	HA
II. Language and Style	3.00	3.60	3.80	3.47	0.4619	HA
III. Content	3.40	3.80	4.00	3.73	0.3464	HA
IV. Test Specifications	4.00	3.67	4.00	3.89	0.1925	HA
V. Test Format and Directions	3.56	3.89	3.78	3.74	0.3208	HA
VI. Test Items in General	3.70	3.80	3.90	3.80	0.1732	HA
VII. Multiple Choice Items	3.50	3.90	3.80	3.73	0.2309	HA
Composite	3.55	3.81	3.85	3.73	0.2700	HA

APPENDIX C: TABLE OF SPECIFICATIONS AND 50-ITEM PRE-/POST-TEST

TABLE OF SPECIFICATIONS IN

SCIENCE GRADE 4 (QUARTER 4)

INSTRUCTION AND DESCRIPTION OF THE SUBJECT	Science education aims to develop scientific literacy among learners that will prepare them to be informed and participative citizens who are able to make judgments and decisions regarding applications of scientific knowledge that may have social, health, or environmental impacts. (K-12 Science Curriculum Guide)
GOAL STATEMENTS	The goal statements of this Pretest are as follows: a. Assess grade four pupils' prior knowledge to the topics to be discussed in the fourth quarter of Academic Year 2021-2022; b. Strategize teaching strategies that will help grade four pupils' attain conceptual knowledge and; c. Determine grade four pupils' misconceptions on the topics to be discussed.
INSTRUCTIONAL OBJECTIVES	The instructional objectives of this Unit Test are as follows: a. C: Analyze daily-life cases anchored on scientific concepts using deductive approach; b. P: Test prior knowledge using Google Forms in a 1-hr synchronous session; c. A: Answer the pretest with honesty and perseverance

Week #	**Learning Competencies**	**Time Frequency (in hrs/**	**Weight % of time frequenc**	**Cognitive Process Domain**												**# of Items**	**Weight % of the # of items**
				R		**U**		**Ap**		**An**		**E**		**C**			
				NI	**PI**	**NI**	**PI**	**NI**	**PI**	**NI**	**PI**	**NI**	**PI**	**NI**	**PI**		

		wk)	**y**														
1	Identify the different physical characteristics of the earth	1	20	2	1-2	2	3-4	2	5-6	2	7-8	2	9-10			10	20
2	Describe how light travels (S4FE-IIIf-g-4) Investigate properties and characteristics of light (S4FE-IIIh-5) Describe ways to protect oneself from exposure to excessive light (S4FE-IIIi-j-6)	1	20	2	11-12	2	13-14	2	15-16	2	17-18	2	19-20			10	20
3	Describe the changes in the position and length of shadows in the	1	20	2	21-22	2	23-24	2	25-26	2	27-28	2	29-30			10	20

	surroundings as the position of the Sun changes (S4ES-IVh-9)																
4	Describe how heat travel (S4FE-IIIf-g-4) Describe ways to protect oneself from exposure to excessive heat (S4FE-IIIi-j-6)	1	20	2	31-32	2	33-34	2	35-36	2	37-38	2	39-40			10	20
5	Practice safety measures in physical activities and proper handling of materials (S4FE-IIIb-c-2)	1	20	2	41-42	2	43-44	2	45-46	2	47-48	2	49-50			10	20
TOTAL		5	100													50	100

VALIDATED TEACHER-MADE TEST

Directions: This test consists of five parts, divided based on the topics oriented to grade four pupils, namely:

1. Earth's Physical Characteristics
2. Properties of Light and How it Travels
3. Changes in the Position and Length of Shadows in the Surroundings as the Position of the Sun Changes
4. Properties of Heat and How it Travels
5. Safety Measures in Physical Activities and Proper Handling of Materials

Each part will have ten items that you will need to answer. Read the following sentences and choose the letter of the best answer.

01. Which planet in the solar system sustains life?

a. Earth
b. Mars
c. Mercury
d. Venus

02. About how much of the Earth is covered in water?

a. 21%
b. 55%
c. 71%
d. 95%

03. Which among the following characteristics is Earth distinctive to other planets?

a. life exists
b. sun gases
c. rocky surfaces
d. poisonous atmosphere

04. What characteristics of the Earth give day and night?

a. orbit
b. rotation
c. revolution
d. solar eclipse

05. Which of the following does NOT describe revolution?

a. use to determine years on a calendar
b. movement of one object around another
c. use to help create Global Positioning System
d. takes a year for earth to move around the sun

06. You wear a pair of sunglasses, a hat, and thin light-colored clothes because you feel that today is very hot as compared to the cold breeze yesterday. Among the physical characteristics of the Earth, what does this scenario refer to?

a. atmosphere
b. climate
c. composition
d. weather

07. A scientist observes that the earth changes its seasons for a period of time. What is the best guess he can make to explain this kind of change? The Earth...

a. is tilted with respect to its orbit around the Sun.
b. has less gravitational pull with respect to the Sun.
c. has a larger gravitational pull with respect to the Sun.
d. rotates and revolves with respect to its orbit around the Sun.

08. A group of friends were fondled on the coming ship along the Manila Bay horizon. They knew that the horizon was a straight line, yet when the ship approached the shore it appeared to be slightly flattened at the poles. With this observation, what could it say about the shape of the Earth?

a. concave
b. flat
c. irregular
d. spherical

09. Read the scenario and supply the most appropriate reason to the student's answer.

Teacher: What do you think will possibly happen if the Earth would be smaller in size?

Student: The Earth will possibly have no moon because of its _____.

Teacher: That is correct!

a. weaker gravitational pull
b. constant volcanic eruptions
c. dangerous meteorite impact
d. inability to create life on Earth

10. Ligaya read aloud her Science book and found out that one of the Earth's characteristics influence time and tides. Mark answered and Ligaya approved. What is Mark's answer?

a. atmosphere
b. moon
c. shape
d. size

11. What form of energy behaves like a wave and a particle?

a. heat
b. light
c. noise
d. sound

12. Which of these is the most important natural source of light?

a. fire
b. sun
c. lightbulbs
d. electric lampshade

13. Light travels in _____.

a. circles
b. straight line
c. waves
d. zigzags

14. The measure of light reflected by an object relies on all of the following **EXCEPT**;

a. color
b. texture
c. material source
d. light

15. Cathy would like to choose from four different colored shirts. She knew that each colored shirt absorbs a different amount of light. Which among the colored shirts should she **NOT** wear on a hot day?

a. black
b. pink
c. white
d. yellow

16. Which among the following scenarios show protecting oneself from exposure to too much light?

a. Charlie stares at the sun all morning and afternoon.
b. Anne watches Netflix for more than eight hours a day.
c. Ben reads his e-book with maximum light in a very dark room.
d. Daniel takes a break every ten minutes away from his computer screen to rest his eyes.

17. Venus experimented to analyze if light always travels in a straight line. She has three cardboards and one lighted candle. Each cardboard has one hole. She aligned the cardboards A and B such that their holes are aligned. She put the lighted candle on the opposite end. However, cardboard C was not aligned to cardboards A and B. Based on this preparation, do you think Venus will see the lighted candle?

a. Yes, because the light coming from the candle is bright.
b. Yes, because the light can still go through cardboards A and B.
c. No, because cardboard C blocks the light that goes through cardboards A and B.
d. No, because the hole of cardboard C is not in place in a straight line with the holes of cardboards A and B.

18. The color red is most used as a danger signal to travelers because it can be seen at long distances. The reason behind this is that the color red has the...

a. least scattered
b. highest energy
c. highest wavelength
d. Both A and B

19. Allen looks at a swimming pool roughly 3-ft deep. He believes that what he sees is false as to how deep the pool is. His brother told him that the light refracts to make the height of the pool look shallow. Is the brother's explanation correct?

a. No. Light passed through the pool's surface, causing the pool's depth to change.
b. No. Light bounces after hitting the pool's surface, causing the pool's depth to change.
c. Yes. Light moves fast when it travels from air into the water, causing the pool's depth to change.
d. Yes. Light slows down when it travels from air into the water, causing the pool's depth to change.

20. "The reason why we cannot see non luminous objects or those objects without natural light is because we cannot project light out from our eyes." Is this sentence scientifically correct?

a. Yes. Light comes out of our eyes because light is made in our optic nerve.
b. Yes. Light comes out of our eyes because light is transmitted from our brain.
c. No. We cannot see non luminous objects because light does not reflect directly from the source to our eyes.

d. No. We cannot see non luminous objects because there is no reflection of natural light to make objects visible.

21. To create a shadow, one must have _____.

a. light source
b. opaque object
c. surface behind the object
d. all of the above

22. Which among the characteristics of shadow is/are correct?

a. depends on the color of the object
b. shows both the outline and the details of the object
c. depends on the distance between the object and the source of light.
d. Both A and B are correct.

23. When the Sun is behind you, your shadow is _____

a. not there
b. behind you
c. in front of you
d. at the side of you

24. What is the size of shadow when an object moves closer to a light source?

a. big
b. flat
c. long
d. small

25. You go outside to see your friend. Both of you face each other and observe that your shadows have different sizes. Yours is longer, while his is short to none. What is the sun's position in relation to your friend's position?

a. above
b. below
c. front
d. back

26. You go out of your house at three in the afternoon, the sun's position is above your head. What will your shadow look like?

a. elongated
b. short
c. small
d. none

27. If your shadow is in front of you, then you will know that the sun will be _____ you.

a. above
b. behind
c. below
d. beside

28. You put an object outside of your house and observe its shadow. In four hours, you have recorded at least four different shadow positions. What do you think is the reason for these changes?

a. Sun's position
b. Earth's motion
c. object's position
d. both A and C

Refer to the table given for numbers 29-30.

Sarah, Cedie, and Jack recorded the length of their shadows every three hours.

	Sarah	Cedie	Jack
09:00 AM	long	long	long
12:00 NN	none	none	short
03:00 PM	long	short	long

29. Based on the table, which among the three persons recorded his/her observation correctly?

a. Sarah
b. Cedie
c. Jack
d. no one

30. Based on the table in number 29, what is the possible position of the Sun in relation to the three persons' position at around 12:00 NN?

a. above
b. behind
c. beside
d. below

31. What is the type of energy that transfers from a higher temperature object to a lower temperature object?

a. heat
b. light
c. nuclear
d. sound

32. Heat transfers from _____ objects to _____ objects.

a. cooler; warmer
b. warmer; cooler
c. metal; nonmetal
d. nonmetal; metal

33. All of the following are insulators **EXCEPT**;

a. paper
b. pencil
c. rubber
d. frying pan

34. Which material will heat travel through the fastest?

a. wood
b. paper
c. metal
d. plastic

35. What type of heat transfer is happening when people sunbathe?

a. refraction
b. radiation
c. conduction
d. convection

36. Which of the following should you do to protect yourself from excessive heat?

a. Apply a little amount of sunscreen
b. Use tanning beds without using sunscreen

c. Apply sunscreen with SPF every two hours
d. Go outside and sunbathe for as long as you want

37. Experts say that during summer, we should drink more frequently because our body functions best within a specific temperature range. If this is the case, then the body will _____.

a. cool off
b. heat more
c. perspire less
d. perspire more

38. Car enthusiasts prevent people from staying or leaving anyone in closed, parked cars because the car is a/an _____.

a. conductor
b. insulator
c. radiator
d. refractor

39. A mother wraps her baby in a thick and heavy blanket during the summer, because she believes that the baby's skin is not yet well-developed. Does the mother protect her baby from excessive heat correctly?

a. Yes, the mother knows best for her child.
b. Yes, the baby should always be wrapped, regardless of the heat.
c. No, the baby cannot tolerate the heat well because his skin is thin and sensitive.
d. The situation lacks information.

40. Channelle vomits and refuses to drink water after a 5-kilometer run. Her friend Jaya is worried as Channelle seems likely to faint. What should Jaya do?

a. Call 911 for an emergency
b. Give Channelle a bottle of soda
c. Give Channelle a bottle of coffee
d. Put a very cold towel on Channelle's neck

41. Which among the following may cut objects?

a. pair of scissors
b. lighted candle
c. fixed wirings
d. spilled milk

42. Which among the following may burn objects?

a. pair of scissors
b. lighted candle
c. fixed wirings
d. spilled milk

For numbers 43 and 46, refer to the list of proper handling and moving materials.

I. always know the emergency contact information
II. always keep the door closed
III. label poisonous and toxic chemicals

IV. avoid lifting materials from the floor or while seated
V. segregate wastes and practice the Reuse, Reduce, and Recycle
VI. apply alcohol to disinfect oneself
VII. put all sharp and blunt objects away
VIII. prevent octopus wiring of technological devices

43. Which among the following suggestions will lead to a greener and cleaner environment?

a. I and II
b. III and IV
c. V and VI
d. VII and VIII

44. Which among the following suggestions will prevent electrical problems?

a. II
b. III
c. VI
d. VIII

45. Your mother bought a silver cleaning liquid and put it beside the water. Which among the following suggestions will prevent toxification?

a. I
b. III
c. V
d. VII

46. Your brother and sisters charge their phones in a single extension. Which among the following suggestions will prevent overheating and explosion?

a. I
b. II
c. VII
d. VIII

47. Chantal washed her hands with soap and water. She wet her hands with clean and running water. Scrubbed for at least 20 seconds. Rinsed and dried using a clean towel. She forgot to do one procedure in proper washing of hands. What do you think is that missing step?

a. Drink water to rejuvenate
b. Apply face powder to feel fresh
c. Do a manicure to add beauty to nails
d. Lather the back of hands, between fingers, and under the nails.

48. Germs and viruses can easily attack one's body when _____.

a. Eyes, nose, and mouth are touch with unwashed hands
b. Prepare or eat food and drinks with unwashed hands
c. Touch a contaminated surface or objects
d. All of the above

49. The floor is flooded because of a typhoon. There is a loose electrical wiring touching the flooded floor. The circuit breaker is far away from you. What should you do?

a. Nothing. Let the flood subside first.
b. Let others pull down the circuit breaker for you.
c. Get a metal rod to pull down the circuit breaker.
d. Pull down the circuit breaker to stop the flow of electricity.

50. Your friend Liza fainted because of suffocation of heat. All of your friends approach Liza to see what happened. What would you do?

a. Join your friends and chat on what happened.
b. Leave the accident and let your friends sort it out.
c. Do a first-aid emergency that you yourself don't know.
d. Tell your classmates to move away from Liza and call 911.

APPENDIX D: 5-ITEM ESSAY AND RUBRIC

This section gave pupils the opportunity to say their position about the sub themes of fourth quarter, namely:

1. Characteristics in relation to Earth's Characteristics
2. Culture in relation to Properties of Light and How it Travels
3. Worldviews in relation to Shadows
4. Science Technology and Society in relation to Properties of Heat and How it Travels
5. Global Understanding and Call for Action in relation to Safety Precautions.

Directions:

1. After watching the video, answer the question in NO LESS THAN 3-SENTENCES. Refer to the rubrics provided.
2. This essay will run through anti-plagiarism software. Make sure to not copy and paste contents from the internet; otherwise, your answer will automatically be zero (0).
3. Your score will be an average of all these criteria with the formula: [(FD+OR+VO+SGMS)/4]

Guide Questions:

1. What are the physical characteristics of the earth that makes it habitable? What do you think would happen if one or more characteristics/ is/are absent?
2. How do the properties of light and its different mediums create cultural diversity among different colors of the people?
3. Why do people have different cultures, beliefs, and values in our natural environment? What makes them create opinions about their community?
4. Why is science and technology important to create a society? Give some examples of how science and technology improve society?
5. Besides the rapid spread of the COVID-19 virus and other environmental issues, one virus is discrimination to other cultural groups. What is your stand/opinion about this? How can a grade four pupil like you invite other people to show kindness rather than discrimination?

	4 PTS.	3 PTS.	2 PTS.	1 PT.
Focus and Details (FD)	There is one clear, well-focused topic. Clear and detailed main idea is present.	There is one clear, well-focused topic. Clear and somewhat supported main idea is present.	There is one clear topic. Main idea is somewhat clear.	There is an unclear topic and main idea.
Organization (OR)	The introduction is inviting, states the main topic,	The introduction states the main topic and	The introduction states the main topic. A	There is an unclear introduction,

	and has an overview of the subtheme. Information is relevant and presented in a logical order. The conclusion is strong.	provides an overview of the subtheme. A conclusion is included.	conclusion is included.	structure, or conclusion.
Voice (VO)	Purpose is very clear and there is strong evidence of attention. There is extensive knowledge and/or experience with the topic.	Purpose of writing is somewhat clear, and there is some evidence of attention. There is knowledge and/or experience with the topic that is/are evident.	Pupil's purpose of writing is somewhat clear, and there is evidence of attention. There is limited knowledge and/or experience with the topic.	The pupil's purpose of writing is unclear.
Sentence Structure, Grammar, Mechanics, and Spelling (SGMS)	Well constructed sentences and varied structure. No grammatical errors and/or spelling.	Most sentences are well constructed and have varied structure. Few grammatical errors and/or spelling but do not interfere with understanding.	Most sentences are well constructed, but they have a similar structure. Several grammatical errors and/or spelling that interferes with understanding.	Sentences sound awkward, are distractingly repetitive, or are difficult to understand. The pupil makes numerous errors in grammar, mechanics, and/or spelling that interferes with understanding.

Adapted from: International Reading Association. (2013). NCTE: Essay Rubric.

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APPENDIX E: ETHICAL CONSIDERATION AND CONSENT FORM

Types of Participants

The researchers chose to select the population of grade four pupils of the Philippine Normal University (Manila) - Institute of Teaching and Learning (ITL). The researchers assume that the respondents are credible, typical, and affirm the core values of the university. Specifically, there are 32 grade four pupils enrolled in the Third Quarter of the Academic Year 2020-2021 under PNU-ITL.

Risk/s to Participants and Researchers

There will be no risks to participants and researchers for the entire research process. As both parties are in a bichronous distance mode of learning, it is expected that they are in a healthy learning environment free from any forms of issues (e.g. medical, physical, mental, etc.). Should there be any concerns on the risks of participants to be enjoined in this research, researchers will bring the issue to the respective authorities of PNU-ITL for possible consultation and remediation of the researchers and participants.

Benefits to the Participants or Community

More than the usual standard of assessing and improving pedagogical techniques, this study is vital to enhance teaching and learning materials that should be effective and efficient in bichronous distance learning.

Confidentiality and Handling of Data

The researchers are committed to respecting data privacy. No other persons will have access to personal information or identification without the participants' consent, parents, and the PNU-ITL authorities unless required by the law. Study numbers will be used for any personal data to ensure anonymity. All information will be kept confidential and stored in a secured flash drive that only the researchers will have access to. No identifying information will be allowed off-site in any form. The researchers will also give consent and assent forms to the participants to ensure researchers and participants that the data to be collected will be used only for the study itself. No further projects will use this data unless required as an extension of the study.

Learners' Consent Form

I acknowledge that the researchers and their partners explained the research study to me. I read the statements in this letter of information and confirmed that they explained the study information and procedures to me during the consent discussion. I had the opportunity to ask questions about the study. I was informed of my right not to consent to participate; my right to withdraw without compromising the research quality. The researchers also explained the potential risks, harms, and discomforts before the examination time. I also understood the benefits (if any) of participating in the research study.

I understand that I have not waived my legal rights nor released the researchers, school, and officials involved in their legal and professional duties. I know that I may ask now, or in the future, any questions we have about the study or the research procedures. I have been assured that records relating to my care will be kept confidential and that no information will be released or printed that would disclose my identity without permission unless required by law. I have been given sufficient time to read and understand the above information. Therefore, I will sign this consent to participate in this research study.

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Researchers' Study Confirmation

We, Teacher Stephen Allen M. Dy and Teacher Clarisa May Gabion, confirm that we have explained the study to the grade 4 pupils and have supplied them with a signed and dated copy of the consent form.

We, Teacher Stephen Allen M. Dy and Teacher Clarisa May Gabion, are the researchers responsible for the conduct of this study at Philippine Normal University (Manila) - Institute of Teaching and Learning.

For further information

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Rubric for Evaluating a Reflection

Criteria		Levels of Performance				Ratings		
		10 *Outstanding*	**7** *Proficient*	**5** *Basic*	**3** *Below Expectations*	Self 30%	Teacher 70%	**Sub-total**
Observations and Insights	50%	Thoughtful observation of details was evident. Synthesis has high degree of insightfulness and clearly connected to real-life situations or previous content.	Substantial details were noted. Insights were insightful and connected to real-life situations or previous content.	Details were noted. Insights were insightful and connected to real-life situations or previous content.	Very simple details were noted with very little or no insights, comments or analysis.	50		
Depth of Reflection	40%	Entries are evident of personal realizations and convictions to improve understanding of oneself and future performance.	Entries are evident of connections to real-life situations and possible actions to improve understanding of oneself and future performance.	Entries are evident of connections to real-life situations but very limited understanding that there are ways to improve of oneself and future performance.	Entries simply presented how the experience relate to real-life situations.	40		
Promptness	10%	The Worksheet and Reflection were submitted before the agreed deadline.	The Worksheet and Reflection were submitted on the agreed deadline.	The Worksheet and Reflection were submitted 1 day after the agreed deadline.	The Worksheet and Reflection were submitted 2 or more days after the agreed deadline.	0		
						90		
TOTAL	100%							
							Equivalent	
Signature of Student								